Data Communication and Computer Networks

MIDSEM PRACTICE PAPER | 30 marks | Suggested time: 90 minutes

Modules 1-2

Module boundaries checked against DCCN - Detailed Syllabus.pdf in Teams. Covers signals, modulation, line coding, network devices, layered architecture, error detection, flow control and Ethernet access. Q1 and Q2 are assigned 6 marks each instead of the original 5 to make a 30-mark practice paper.

Answer all questions. Show working and draw labelled diagrams where appropriate. Marks and duration are a practice format, not a confirmed university blueprint.

Question 1 [6 marks]

Compare Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) in terms of how each modifies the carrier signal, their bandwidth efficiency, and susceptibility to noise. Analyse which technique is most suitable for transmission over a noisy telephone channel and justify your choice.

Source: DCCN-R | original Q1A

Question 2 [6 marks]

Define data rate and bandwidth in a composite signal. Approximate a square wave by
$s(t) = (4/\pi)[\sin(2\pi \times 2 \times 10^6 t) + (1/3)\sin(2\pi \times 6 \times 10^6 t) + (1/5)\sin(2\pi \times 10 \times 10^6 t)]$.
Calculate the data rate and bandwidth. Triple the frequency and calculate the new data rate and bandwidth. Explain how bandwidth affects data rate.

Source: DCCN-M | original Q1A

Question 3 [2 marks]

For the given binary data 101000101010, draw the following line coding scheme. (Assume previous pulse was negative) i. Differential Manchester ii.Bipolar AMI

Source: DCCN-R | original Q3C

Question 4 [2 marks]

For the given binary data 111000101010, draw the following line coding scheme. (Assume previous pulse was negative) i) Manchester ii) Bipolar AMI

Source: DCCN-M | original Q3C

Question 5 [2 marks]

A system has an n-layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers, an h-byte header is added. What fraction of the network bandwidth is filled with headers?

Source: DCCN-R | original Q1C

Question 6 [2 marks]

A company network consists of three departments connected as follows: Sales and HR are connected through a hub, HR and IT are connected through a switch, and the switch is connected to a router for internet access. An employee in Sales sends a file to an employee in IT. Analyse how the hub, switch, and router handle this transmission.

Source: DCCN-M | original Q1C

Question 7 [3 marks]

Apply the Cyclic Redundancy Check (CRC) method to the data D = 1010001101 using the generator polynomial P = 11011. Determine the transmitted codeword after performing the CRC encoding process.

Source: DCCN-M | original Q1B

Question 8 [3 marks]

Assume that the sender's window size is 3. If we have to send 10 frames in total, and the channel of transmission is such that every 5th transmission is lost (but no ACKs), how many data transmissions (including both original transmissions and any necessary retransmissions) does the sender make in total if:- i. The go-back-N protocol is followed ii. The selective repeat protocol is followed Also, calculate the ratio of the effective number of frames sent to the total number of frames sent in each of the above cases.

Source: DCCN-R | original Q1B

Question 9 [2 marks]

Calculate the total time required to transmit 1,000,000 bits using the Stop-and-Wait ARQ protocol when each frame carries 2000 bits, the sender and receiver are 5000 km apart, and the propagation speed is 2 × 10^8 m/s. Ignore transmission, waiting, and processing delays, and assume no frames are lost or damaged

Source: DCCN-R | original Q4C

Question 10 [2 marks]

An Ethernet segment has transmission speed 10^8 bits/second and maximum length 1500 metres. If propagation speed is 3×10^8 metres/second, calculate the minimum frame size in bits required for collision detection.

Source: DCCN-M | original Q4C

Sources and selection notes

Working interpretation: NSM means end-semester. This has not yet been confirmed. Questions come only from the local semester-3 CSE papers, using the exact current subject titles. R = regular; M = makeup. Wording is lightly edited, tables transcribed, original question references retained. No new question scenarios were invented.

DCCN-R: 2025-26 Regular - CSS_2102-Data Communication and Computer Networks

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DCCN-M: 2025-26 Makeup - CSS_2102-Data Communication and Computer Networks

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